

BlinkChat: Blink-Augmented Prompting for Adaptive Human-LLM Interaction

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ABSTRACT

Current Large Language Models (LLMs) rely on explicit and user-initiated prompts for contextual adaptation, leaving passive user-state signals largely unexplored in interaction design. In contrast, using involuntary physiological signals as control channel for adaptive human-LLM interaction is relatively underexplored. In this extended abstract, we explore this idea by choosing human blinking as an involuntary physiological signal. We introduce *BlinkChat*, a prompting framework that encodes eye-blink activity as prompt. By locally integrating control tokens directly into the input stream, blink events are detected locally and injected during typing. This design does not modify any current user-LLM interaction dynamic nor transmit raw biometric data to the remote server. In terms of application, we consider blink frequency as a coarse proxy for cognitive strain, and a system-level prompt instructs the LLM to interpret and adjust response verbosity accordingly while preserving correctness. We evaluate this approach using 400 long-form prompts. Our simulation result shows task-dependent reductions in response length by up to 31.5% at high blink rate (30 blink per minute), with compression effects strongest for short-answer factual tasks and attenuated for reasoning-intensive domains.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design.**

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1 INTRODUCTION

Large Language Models (LLMs) are rapidly transitioning into interactive components embedded in productivity tools [11], educational systems [15], and autonomous agents [14]. Despite substantial progress in reasoning and controllable generation, the interaction paradigm between humans and LLMs remains **explicit and user-initiated** [16]. Users must explicitly provide prompts and actively share contexts to steer the system. These mechanisms require user awareness or deliberate configuration. In contrast, using involuntary physiological signals as control channel for adaptive human-LLM interaction is relatively underexplored [4]. Many signals

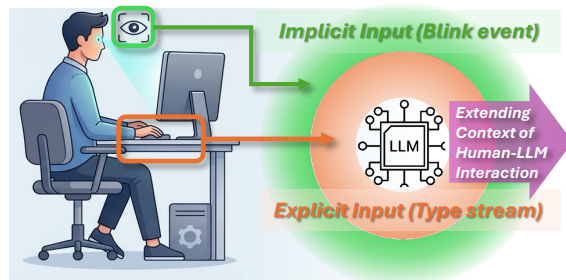


Figure 1: BlinkChat: Augment Human Blink into current Human-LLM Interaction

reflecting user state, such as **eye blink frequency** are inherently passive: they can be detected continuously without requiring conscious user input. We argue that incorporating passively sensed signals into the prompting layer represents a distinct and underexplored design space.

Blink rate has been shown to correlate with cognitive load, stress, and attentional variability [6]. Elevated blink rates can be used as a proxy for increased cognitive strain [1, 3]. We hypothesize that the user may benefit from more concise responses at high cognitive strain, while normal blink conditions preserve standard output behavior. Importantly, we do not attempt to infer fine-grained emotional states. Instead, we treat blink frequency as a coarse-grained control variable to regulate response verbosity. In this study, we are interested in the research questions: (1) *How to augment human blink information with human-LLM interaction?* (2) *Does blink-augmented prompting produce measurable cognitive adaptive responses?*

To answer these questions, we design a blink-conditioned prompting framework that encodes blink frequency into the system-level prompt without modifying the underlying model architecture. We simulate 5 different blink-frequency conditions and evaluate their effect on model behavior using 400 representative user prompts spanning common instructional tasks. We measure adaptive response characteristics in terms of output verbosity and assess task success relative to a baseline without blink conditioning. This controlled design allows us to isolate the effect of passive blink augmentation on response adaptation while preserving answer quality. We compared the output length reduced by up to 31.5% as compared to the baseline.

2 SYSTEM OVERVIEW

Local Involuntary Eye Blink Detection Blink detection is performed entirely on-device using the MediaPipe Face Landmarker



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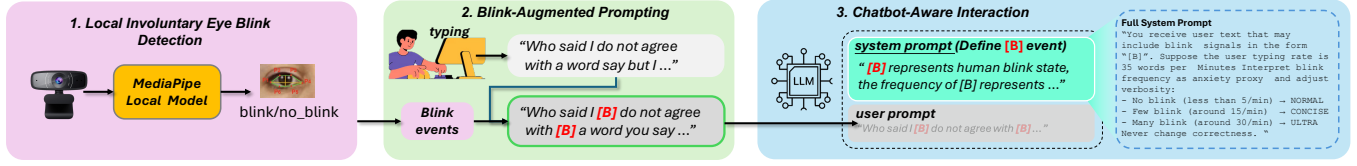


Figure 2: *BlinkChat* System Overview

model [7] (Figure 2.1). The model produces 3-D landmark coordinates, and blink is determined on the Eye Aspect Ratio (EAR) metric [13]. EAR approaches 0 when the eye is fully closed and is approximately 0.25–0.35 during normal open-eyed gaze.

Blink-Augmented Prompting When a blink onset event is detected, the system inserts the annotation token [B] at the nearest word boundary (Figure 2.2). The marker [B] is chosen as it is distinct from natural-language words, and short enough that inserting many copies without dramatically inflating the input tokens.

Chatbot-Aware Interaction We use the system prompt to achieve the blink-awareness during the chatbot interaction to allow the model to pay attention to the customized token [B]. The full prompt that learn the blink-rate-to-verbosity mapping is displayed as in Figure 2.3.

3 EXPERIMENT

Datasets Four publicly available datasets (NaturalQuestions [12], GSM8K [5], TriviaQA[8], HumanEval[2]) were chosen to span a range of prompt variation and complexity. For each data set, to maximize the variance of blink-signal (longer sentences allow more blink events), the 100 longest prompts per data set were selected.

Blink Signal Simulation via Poisson Process The simulation models blink events arrive according to a homogeneous Poisson process. Suppose user typing rate is $wpm = 35$ words/minute [9], blink event can be modeled as Poisson blink process [10]. Blink inter-arrival times are drawn from an exponential distribution $\Delta t_k = -\ln(U_k)/\lambda$, $U_k \sim \text{Uniform}(0, 1)$ with rate $\lambda = N_{bpm}/60$ events/second. Five blink-rate conditions were generated per prompt sample with six variants $N_{bpm} = 0, 5, 10, 15, 30$, and 35 blinks/minute.

LLM Evaluation All evaluations used Qwen/Qwen2.5-7B-Instruct loaded in bfloat16 on a single NVIDIA GeForce RTX 3090 GPU. Generation settings control temperature as 0.7 and top_p as 0.9 and remain constant throughout.

Verbosity Metric For each prompt, the model was queried 5 independent trials. For each trial, both word count and generated token count were recorded. The length averages were then aggregated across all 100 samples to produce condition-level means and standard deviations.

4 RESULT

Figure 3 displays the mean word and token counts in all four benchmark datasets. NaturalQuestions exhibits the clearest monotonic verbosity reduction (31.5% at 30 blinks/min). TriviaQA shows a consistent but shallower trend: word count decreases from 89.9

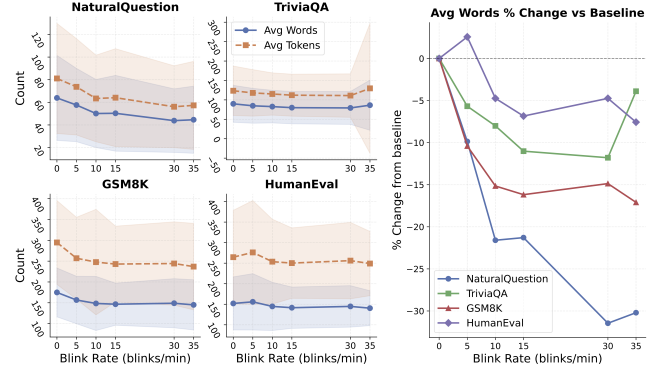


Figure 3: Average Response Length (word count and token count) vs Blink Rate for Four Datasets

to 79.3 at 30 blinks/min (-11%). GSM8K (grade-school mathematics) produces an initial, meaningful compression from 174.9 words (baseline) to 156.7 at 5 blinks/min (-10.4%), but the trajectory quickly plateaus. HumanEval (code generation) shows the most muted response. The degree of verbosity aligns with task complexity: short-answer factual queries show strong monotonic reductions, while reasoning-heavy and code-generation tasks maintain a verbosity floor to preserve correctness.

5 CONCLUSION

We presented *BlinkChat* and demonstrated that, by encoding blink events as lightweight inline tokens, large language models exhibit measurable cognitive adaptation in response length. These results suggest that passive and implicit physiological signals can function as a wider channel for human-LLM interaction. Future work may extend this framework to real user studies, multimodal signals, and more fine-grained adaptive policies, advancing toward truly context-aware and cognitively adaptive LLM systems.

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